

The Effect of Shape Differences on Object Impressions and Their Relation to Weight Perception

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Abstract

This study focuses on the impression differences caused by shape and investigates the relationship with weight perception. There has been a lot of attention to studies on the illusion caused by the manipulation of visual information using virtual reality (VR) and augmented reality (AR) in recent years. It is known that size and texture affect weight perception. On the other hand, few studies have explored the relation between weight perception and shape, which is one of object properties. Besides, its factors and mechanisms are unclear. Therefore, this study investigates the effect of impression by each shape on the weight perception using a VR environment which allows object shapes to be easily changed.

ACM Reference Format:

Kosei Tomioka, Naoki Hashimoto, and Mie Sato. 2025. The Effect of Shape Differences on Object Impressions and Their Relation to Weight Perception. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771437>

1 Introduction

There has been much attention focused in recent years on the phenomenon of optical illusions caused by changes in visual information in VR and AR. In particular, there are known phenomena such as Material-Weight Illusion (MWI)[Buckingham et al. 2010] and Size-Weight Illusion (SWI)[Nicolas et al. 2012], in which the perception of weight changes depending on differences in size and texture of the same mass. This has been followed by research on the influence of shape on the perception of weight, or Shape-Weight Illusion.

In the real space study[Ryu et al. 2022], spheres were perceived as the heaviest, followed by cubes and tetrahedrons. In addition, it reported differences in perceived surface area depending on shape, but also suggested that surface area by itself cannot entirely explain the weight perception.

The relationship between shape and weight perception has not been fully clarified, and there are few studies on this topic. Therefore, we focused on the influence of the different impressions given by shape on weight perception in this study. Then, we conduct

experiments using a method in which virtual shapes are overlaid on real objects in a VR space.

2 Experiment

2.1 Experimental stimuli

A total of nine virtual objects were prepared. The basic shape is a cube, which has been used in previous studies, and eight other shapes that showed significant differences in impression based on preliminary experiments. Figure 1 shows the experimental stimuli. These shapes were created by combining 27 small cubes, which were cut into three equal parts in all three directions (width, height, and depth) of the cube to ensure equal volume.

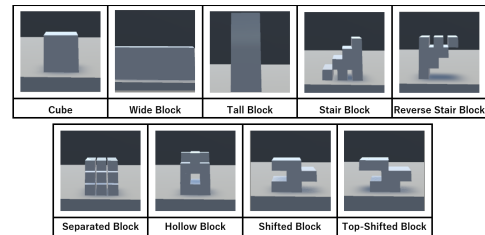


Figure 1: Virtual Objects Used in the Experiment

2.2 Experimental environment

The VR environment was developed using Unity (2019.4.28f1) as the development environment. This experiment used an HMD (HTC Inc., VIVE Pro), a motion capture system (OptiTrack, PrimeX13, Motive2.2.0), and same box-shaped objects with a handle which weights 0.8 kg, 1.0 kg, and 1.2 kg. These objects were selected to standardize the ways of holding them, in addition to limiting the size of various virtual objects. The subjects viewed virtual objects that imitated the objects to grasp in the experiment, as well as various objects of different shapes placed at the center of the bottom surface of the virtual objects.

2.3 Experimental procedure

This experiment was conducted in the following order: a pre-questionnaire, a shape evaluation task, an estimate center of gravity task, a weight comparison task, and a post-evaluation.

In the shape evaluation task, we investigated what impressions subjects had about the presented stimuli before grasping them. In this task, nine types of virtual objects were presented to the subjects in random order. The subjects observed the virtual objects and then answered questions Q1 to Q10 on a 7-point Likert scale. For Q1 to Q9, “Strongly disagree” was -3, “Neither agree nor disagree” was 0, and “Strongly agree” was +3. For Q10, “Small” was -3 and

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771437>

“Large” was +3. These questions were selected through preliminary experiments to more precisely assess impressions.

As for Q1-Q9, the question sentences are “Does the virtual object look like (Q1-Q9)?” (Q1: it would fall over easily, Q2: it has a wide contact surface, Q3: elastic, Q4: streamlined, Q5: complex in shape, Q6: symmetrical, Q7: soft, Q8: luxury, and Q9: artificially designed). Q10 is “Compared to the standard cube, how big is the volume?”

In the estimate center of gravity task, we investigated the degree of error in the subjects’ estimation of the center of gravity, which has been suggested to be related to weight perception, compared to its actual position.

The weight comparison task aimed to obtain the degree of weight illusion for each shape. In this task, the subject was first instructed to extend his arm diagonally 45 degrees downward in front of himself (initial position). Then, on cue, the subject grasped the grasped object, lifted it to the same height as his shoulder, and then returned it to the initial position. Weight comparisons were made according to the following procedure.

- (1) The subject performed the lifting motion while looking at a cube-shaped virtual object as a reference.
- (2) The subject performed the lifting motion while looking at the virtual object that was the experimental stimulus.
- (3) The subject answers the weight of (2) when the weight of (1) is set to 1.0.

Figure 2 shows the state of the experiment during the weight comparison task and the state of the VR space that the subject sees through the HMD. Although the trials analyzed were all 1.0 kg, dummy trials were conducted with different weights to avoid realizing that the weight was always the same. The number of trials was 22, including 9 types of experimental stimuli, rounded corners or not, and 4 dummy weight trials, with steps (1) to (3) constituting one trial. These were conducted in random order.

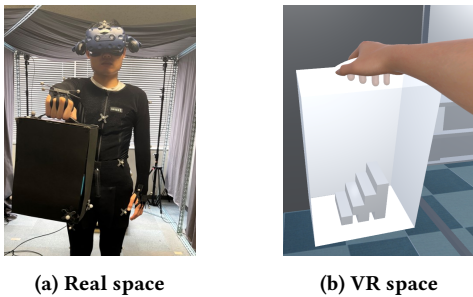


Figure 2: The Scene During the Weight Comparison Task

3 Results

In this experiment, 20 male subjects between the ages of 18 and 24 participated. However, one subject was dropped because he did not give the right answer in a trial where it was clear that there was a difference in weight. Therefore, we used data from 19 subjects for the analysis. First, we examined the relationship between each question item and the weight illusion during the shape evaluation task. We calculated Spearman’s rank correlation coefficient and found that the effects were small in all cases. However, a positive correlation was observed in Q9 ($r = .20, p = .03$) and a negative correlation was observed in Q4 ($r = -.24, p = .03$) and Q7 ($r =$

$-.15, p = .02$). This suggests that smooth, continuous shapes and soft impressions may be perceived as lighter than they actually are.

Next, Figure 3 shows the results of the weight illusion in the weight comparison task. Since a significant difference was shown by the Friedman test ($p = .050$), we performed a multiple comparison test using the Holm method. The results of the multiple comparison test showed a significant difference ($p = .03$) between the cube and hollow block, and a marginally significant difference ($p = .08$) between the reverse stair block and hollow block.

Here, we focused on the impressions that were particularly different among these three shapes. The results showed significant differences between the reverse stair block, cube, and hollow block in Q1, Q2, Q5, and Q6. In addition, significant differences were observed between the cube shape and hollow shape in Q9, and a marginally significant difference was observed in Q8.

This suggests that objects that appear to be less likely to fall over and their shapes are easier to identify may be perceived as lighter. It is also interesting to consider that artificially created impressions and luxury impressions may be perceived as heavier. Furthermore, it is possible that there is a negative relationship between the difficulty of estimating the center of gravity and the ease of identifying shapes. It is worth considering whether it is possible to change these impressions so that they are perceived as lighter.

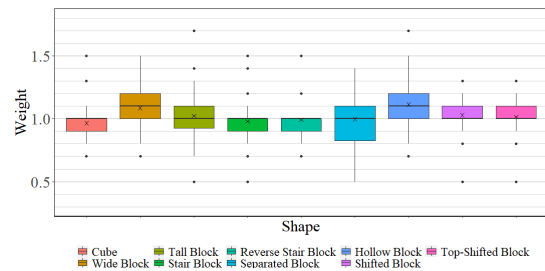


Figure 3: Amount of Weight Illusion for Each Shape

4 Conclusion

This study focused on object impressions as a factor in the unclear influence of shape differences on perceived weight. As a result, it was suggested that the impression of an object may be related to changes in weight perception, whereas the effect of shape differences on perceived weight is limited.

In the future, we will conduct systematic studies to verify the relationship between weight perception and a wider variety of shape and impression factors. We also plan to apply these findings to studies on designs that make objects appear lighter based on their visual impressions, as well as studies on the perception of objects in VR/AR environments.

This work was supported by JSPS KAKENHI Grant Number JP23K11195.

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